KARTHIK K

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Summary

Computer Science undergraduate specializing in Machine Learning, RAG systems, and backend engineering. Built and deployed an end-to-end RAG pipeline with hybrid retrieval and evaluation-driven optimization. Strong DSA foundation (300+ LeetCode problems solved).

Education

Sir M. Visvesvaraya Institute of Technology, Bengaluru, India *2023 – present*
Bachelor of Engineering in Computer Science, CGPA: 8.12/10

Excellent PU College, Udupi, Karnataka *2023*
Pre-University (PCMC), Percentage: 89.3%

Technical Skills

Languages: Python, Java, SQL
LLMs & RAG: RAG Pipelines, FAISS, BM25, Hybrid Search, Cross-Encoder Reranking, Prompt Engineering, Hugging Face Transformers, Groq API
Machine Learning / DL: Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, CNN, Feature Engineering, Model Evaluation
Backend & Tools: FastAPI, Flask, REST APIs, Docker, Git, GitHub Actions, MLflow, CI/CD
Core CS: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks

Projects

Document Q&A RAG Chatbot *Aug – Oct 2026*
Python, FastAPI, FAISS, BM25, Groq/Llama-3.3, Docker

- Built a RAG chatbot with hybrid retrieval (FAISS + BM25) and cross-encoder reranking, improving Hit Rate@5 from 72.2% to 100% on an 18-question evaluation set.
- Improved answer faithfulness from 70% to 90% via prompt engineering, validated using an LLM-as-judge evaluation pipeline.
- Built a domain-agnostic ingestion pipeline supporting any uploaded PDF; deployed as a Dockerized FastAPI service with a custom web UI.
- **GitHub:** `https://github.com/Karthik-kulal8050/rag-chatbot`

Credit Card Fraud Detection *Jul – Aug 2026*
Python, Scikit-learn, SMOTE, FastAPI, Docker, MLflow

- Built a fraud detection system using Random Forest, SMOTE, and threshold tuning for highly imbalanced transaction data.
- Deployed a FastAPI REST API with Docker and MLflow experiment tracking; automated testing via GitHub Actions CI/CD.
- **GitHub:** `https://github.com/Karthik-kulal8050/Credit-card-fraud-Detection`

Brain Tumor MRI Classification *Jun – Jul 2026*
Python, TensorFlow, Keras, MobileNetV2, Streamlit

- Built a four-class MRI classifier using MobileNetV2 transfer learning and fine-tuning for brain tumor classification. Achieved **91.5% test accuracy** and **0.92 macro F1-score** on 1,600 test images and deployed an interactive Streamlit application.
- **GitHub:** `https://github.com/Karthik-kulal8050/Brain-Tumor-Classification`

YOLO Object Detection System *May – Jun 2026*
Python, YOLO, OpenCV, Computer Vision

- Developed an object detection system supporting image, video, and webcam inputs with real-time bounding boxes, class labels, confidence scores, and object counting.
- Implemented configurable confidence thresholds and class filtering for flexible object detection and inference.
- **GitHub:** `https://github.com/Karthik-kulal8050/Object-Detection`

Achievements & Certifications

- Solved 300+ DSA problems on LeetCode.
- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp – Udemy (2025)
- Java Data Structures & Algorithms + LeetCode Exercises – Udemy (2024)